Automatic Furniture Generation Using Pose Priors

A Seminar / R&D Report

by

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Abstract

With the advancement of 3D modelling tools and techniques, there is a growing need for automatic algorithms that can assist a designer in quickly prototyping an object (furniture) model. Related work in this area, has primarily been data-driven, in which people have attempted modifying or assembling existing objects by using different priors on the scene, or objects in the scene. Our work, is motivated by the observation that, the shape of a particular furniture is a function of *comfort* of the human posture that it intends to support. In this project, we offer a novel algorithm for automatically synthesizing a furniture to support a particular human posture. Given a human skeleton pose, we try to estimate the torques, the joints apply internally so as to prevent the skeleton from collapsing, and also at rest. We require the joint torque estimates so as to estimate the sampling density of the contact points on the furniture. This sampling density mimics or simulates a real human body in a particular pose, which needs to be supported by raising a height field around it. Later, we solve an optimization problem which raises the height field, eventually minimizing the joint torque values. We find the estimates by considering the free body diagrams of various parts of the skeleton and applying Newton's Second Law and its derivatives repeatedly on them.

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Chapter 1

Introduction

In this report, we have done an extensive review of the existing literature on *affordance analysis*, *human interaction priors*, *scene design* and *object creation* which are pretty emerging topics in the field of computer vision and graphics. This allows us to understand the functions that a particular object provides to a human actor, with direct application in human computer interaction. Studying affordance of an object allows a robot gain a functional understanding of a particular object, and the environment as a whole, which is often not evident from geometric structure.

"Affordances provide strong clues to the operations of things. Plates are for pushing. Knobs are for turning. Slots are for inserting things into. Balls are for throwing or bouncing. When affordances are taken advantage of, the user knows what to do just by looking: no picture, label, or instruction needed."

- (Norman, 1988)

Another concept which is surveyed is *form follows function*. This is a well-defined concept in modern architecture, and it says that the design of an object should purely depend on the functions that it intends to support. These two primary concepts motivate us to devise a new way of looking at the problem of object creation without using a data-driven model. This might have application in sports and medical equipments, e.g., one might want to design a comfortable seat for a physical disabled person. In another scenario, we might want to design gym equipments that builds stress on targeted muscles.

The rest of the report is organised into three primary sections. In the first part, we discuss, existing literature on the related topics and try to develop an understanding of the state-of-the-art approaches to common problems. We next, present our work with a quick introduction and motivation, and discuss the various technicalities that we have adopted in our approach. The final section contains results that we obtained through our simulations

and experiments. We also discuss the problems that we encountered and why a particular solution was adopted. This is followed by a discussion on the future goals of the project.

Chapter 2

Literature Survey

2.1 Function is Defined by Human Interaction

Affordance analysis is the broad area of Computer Vision and Graphics, where the primary aim is to study, how different objects interact with each other in a particular scene. In this paper (Grabner *et al.*, 2011), the researchers have presented an elegant way of grouping objects that have similar functionality by modeling an imaginary "actor" trying to perform a particular action, and then group objects that allow the actor to interact in the similar way. The broad idea is, that functionality or the actions that a particular object supports, is an indication of its visual appearance. Therefore, this can be used as a cue complimentary to the visual features, to learn and group similar objects in a scene. In addition to this, it can also be used to search the best object that supports a particular action. The most intriguing part of affordance detection is that it requires fewer examples than that required in learning visual features. To understand this with an example, consider the class of furniture that support a sitting posture viz chairs, sofas, benches etc. All these variants have a similar prototype in which there is a handle to support the arm, a seat to support the back and a back rest to support the spine. This crude framework, more or less supports the entire human body and prevents the skeleton structure from collapsing.

2.1.1 Affordance Detection

This work is primarily centered around functionality detection rather than object detection. Specifically, this work assumes that the interest lies in detecting sitting positions in a scene rather than finding objects like chair, sofa etc. Therefore, the approach involves measuring the affordance of a particular part of human body towards the part of the object that can support the particular action. For example: A person can sit on a table, but sitting

on a table is quite uncomfortable as it strains the back muscles. So, an obvious choice for people would be to rest on a chair if given a choice.

2.1.2 Actor-Object Interaction: A General Idea

Affordance is modelled as the interaction between a virtual actor and the 3D scene. Here, the fundamental assumption that drives the process is that the affordance is related only to certain key poses that an actor performs. For example: A person may want to sit, or sleep, or lie on a recliner etc. This assumption allows for human interaction with the objects. However, this can be easily extended to allow object-object interaction, where an object interacts with another objects, for example: a key interacts with a lock. A table affords a laptop to be kept on top of it. The rudimentary idea behind the algorithm is to formulate a probabilistic model defined on the actor's shape that can encode all the variations in the object shape and style.

2.1.3 Mathematical Formulation

An object in the scene is represented as a 3D mesh, denoted as M_i^{object}, and the model of the actor is also represented as a 3D triangle mesh, denoted as M_i^{actor}. In order to make the process efficient, it is required to reduce the number of poses to a certain number of key poses, and therefore, a clustering of the poses are done to determine a mean key pose $\overline{M}_i^{actor}$. A subtle point that could be interesting to experiment is clustering on the basis of some varying weights. This is for answering questions like, "Can I lie down on a wide chair, even though it is uncomfortable?". So, the crux is, can we support parts of the human body with the object? This is relevant to the project we are working on. In future, we want to synthesize a furniture model that can support multiple postures of a human skeleton.

The problem of detecting the functionality of a previously unobserved object, M_i^{object} is formulated as a probability estimation problem, namely estimating the conditionaly probability $p(T|M^{object})$ defined over all transformations T of the key pose model M^{actor} into the scene M^{object} (Grabner *et al.*, 2011). Hence, the final model becomes,

$$p(T|M^{object}) \propto \left(\prod_{j=1}^{|V|} p_j^{dist}(D(TV^j)) \right)^{\frac{1}{|V|}} \cdot \left(\prod_{j=1}^{|T|} p_j^{inter}(I_T(T^l)) \right)^{\frac{1}{|T|}} \tag{2.1}$$

where, $I_T(T^l) = 1$ if the triangle T^l intersects with a triangle of the scene after applying the transformation $\mathbf{T}$ to the mesh, otherwise $I_T(T^l) = 0$.

The probability $p_j^{dist}(d)$ is calculated by reconstructing the the underlying probability of a distance field of a vertex V_j using a kernel density estimator with a Gaussian Kernel,

where 'n' is the number of training examples. And the probability, $p_l^{inter}(l)$ is calculating the mean of intersections of a particular triangular mesh of the actor to any of the meshes of the object.

The intuition behind this formulation is that we learn a probability of the correspondences between parts of the human body with the objects. The probability value is evaluated densely for a particular scene and then determined if a particular human pose can fit on the object model.

2.1.4 Implications

This work introduces us to the very innovative way of classifying objects in scene without featuring mapping the actual surface geometry rather using a human actor as a proxy for the geometric understanding of the surface. This method gave faster results than native approaches of object classification using 3D feature vectors. In addition to this, this model, not only predicts the function that the object supports, but also, how the object can be used by the actor. However, the current implementation of this approach relies only on geometric stability. An extension to this idea, motivates us to approach the problem in a different way by modeling stress diagrams and physical stability relevant to geometry, and subsequently using them for object-affordance prediction.

2.2 Fine-grained Inference of How Humans Interact with Shapes, as a Way to Characterize Shapes

Representation of shapes based on their semantic features has been very relevant for salience estimation, shape segmentation, feature detection, symmetry analysis, and surface correspondence. Algorithms for the above methods have in general used geometric analysis, using computation of global shape properties, extract part structures from them and detect local shape features. Majority of the man-made shapes are designed to be used by/ interacted with humans. This motivates us to leverage the prediction of object affordances during shape analysis. Shape2Pose is a new tool proposed by (Kim *et al.*, 2014) which uses the idea that, knowing how a human interacts with a given shape actually lets us know about the semantics of the shape. The main challenge however in doing so is to predict the most probable pose a human takes while interacting with the object. An efficient algorithm for predicting the pose by exploring the combinatorial space of contact assignments along with a pre-computed probability distribution for body parts.

2.2.1 Algorithm Overview

An affordance model in the form of an objective/energy function characterizing the compatibility of a pose with a 3d model is obtained by training over pairs of 3d models and poses associated with them.The objective functions contains different terms for minimizing different energies like:

- **[Local]** Given assignment of contact points to body parts, separation between them should be minimized
- **[Local]** Feature compatibility energy at each candidate contact point assigned to a body part
- **[Global]** Pose plausibility energy favouring common feasible poses
- **[Global]** Intersection energy for penalizing intersection with surface
- **[Global]** Symmetry energy to see that the object and the pose symmetries are in alignment

The feature compatibility energy is modeled by doing a regression on different training signals for each part on all training shapes along with the geometric features at the sample points of the training signals. The pose plausibility energy is modeled by initially clustering the poses and by learning mean joint angles and simultaneously variances(assuming Gaussian) at the same time penalizing the poses which deviate from left-right symmetry at symmetric joints. The symmetry energy is seen as the deviations from the locally prominent symmetric planes for each part and also penalize improbable separations from the surface symmetry center to the human symmetry center by modeling them as a Gaussian and penalizing the normalized deviation from mean.

Using this affordance model, given an input novel shape an optimal pose is predicted by simultaneously sampling from the likely contact point distributions and the likely pose joint distributions and aligning them by storing the probabilities of being being reached by a body part at points in euclidean space where we also have the distribution of contact points. We align them by rigid transformations and obtain the peaks in the joint probability and through inverse kinematics solve for the exact body pose.

2.2.2 Results and Applications

This method gave more accuracy for many classes of objects when compared to the existing methods to predict a pose like Shape Matching, Rigid Pose, Local Matching.

For forming sparse correspondences between shapes, this method is very useful as the contact points for each body part form correspondences. Also we can use Shape2Pose for Salience Estimation on surfaces which is human-centric i.e salient to human interactions. The problem of view selection can be solved for many scenes by actually fitting a human pose to the scene and selecting the view direction that maximises the sum of human-centric salience over the visible surface area. The problem of shape retrieval is a very vast one in terms of the adjectives that determines the shapes to be retrieved. This method retrieves shapes on semantic similarities which may not be retrieved by standard techniques using the global shape properties.

2.3 New Objects as Ergonomic Deformations of Existing Objects

The ideas discussed in sections 2.1 and 2.2 provide us with two important concepts.

- The functional sense of an object is defined by how humans interact with the object:
- Human interaction can be used as a shape descriptor for object classification and fine grained inference about the appearance of an object.

This idea of human interaction with objects can be exploited further and used to create objects that are small variations of the existing objects. (Zheng *et al.*, 2016) addresses this problem and attempts to solve it by imposing ergonomic constraints on the objects with the help of a human actor, and then making small (local) deformations on the shape to fir the pose. The deformations are subjected to constraints and an iterative algorithm is employed to keep track of the constraints by revisiting the set of constraints again and again. Basically, the work presents a novel reshaping algorithm that links ergonomic considerations to shape reshaping that enables the user to do personal customization. In addition to this, the reshaped object is used for a shape classification setup by calculating a metric, deformation energy which and ranking objects in the database based on that.

2.3.1 Ergonomic Guidelines

Through years of research and painful experiences, a set of ergonomic guidelines have been set. These guidelines are important in setting up the constraints on the shape. Also, following these guidelines ensure the considerable minimization of bio-mechanical stress. The guidelines are summarized in the paper (Zheng *et al.*, 2016) as follows:

- *Chair seats should have correct height* to allow both feet to be fully supported (e.g., by the ground). A chair too high creates undue pressure on at the knee/thigh; while, if it is too short forces the knee to be higher than the hip sockets.
- *Width and depth of chair seats* should conform to the users dimensions. Specifically, while the width is dictated by the avatar's waistline, the dept is dictated by the length of the avatar's thigh bones.
- *Flat un-contoured seats* are preferred to discourage a slouched or C-shaped posture.
- *Lumbar support* by providing low- or mid-back support can help hold good posture and prevent pain to the spine and neck.
- *Head support*, if provided, can help ease stress for the neck muscles and provide support for seating over extended periods.
- *Arm rests*, provide support for reading, typing, painting, and similar activities.

2.3.2 Algorithm Overview

The input to the algorithm is the avatar and the annotated human pose. Then the ergonomic guidelines are mapped to the input shape geometry. The constraints are then integrated with the contact based deformation paradigm to reshape the input object to adhere to the avatar's pose. The reshaping algorithm is designed as a two-step process, in which the first step involves representing the input shape as a graph, where the nodes are the components of the object, and the edges denote the spatial relation between the components of the object. For example: Since, if a chair backrest is in contact with the armrest, then there exists an edge between the nodes "armrest" and "backrest". This is helpful in remembering the overall geometric structure of the object which is to be preserved during the deformation steps. The identification of the ergonomic constraints by converting the contact information into geometric relations. For example: The seat width is determined by the hip width of the human body. The geometric constraints are directly enforced using this contact information, for example, the contact information between the hip and the chair seat top face will lead a particular value of the seat height "h".

In every iteration, these geometric constraints are automatically updated and the shape is steadily deformed to conform to the human posture. An important point to note here is that, the guidelines are primarily based on sitting posture. This involves a restrictive way of designing objects that support a particular posture. Alternatively, one might

want to experiment by modelling the stress relations in the human body and creating geometric constraints while trying to optimize the bio-mechanical stress values at the muscles and the joints. In order to smoothly drive the deformation process, the shape is represented by a proxy, which is essentially a set of cuboids and cylinders that approximate the basic shape. Another point of consideration is that, while the deformation step proceeds, it needs to be ensured that the spatial relations, the overall geometry is preserved and unnecessary deformations are reduced. This is done by adjusting the transformation matrix, i.e. for cylindrical shapes, we enforce uniform scales along the two non-principal axes.

2.3.3 Making Better Furniture

This method exploits ergonomics to deform the shapes to fit a particular human posture. However, the ergonomics derived from the human avatar is not accurate but crude, as a result of which the output shapes are rough. An interesting direction to explore might be to model the human body accurately and deform the shapes into production ready models. In addition to this, the method assumes that the parts of the object is clearly segmented and is annotated properly. A futuristic goal in this work is to experiment by considering other human body stress models, and how the stress distributes on the introduction of a soft cushion as supports for the regions where the bio-mechanical stress is considerably high (e.g., backrest, seat).

2.4 Modeling Functionality by Learning How Regions of Shapes Interact with Each Other

The ideas discussed in the sections 2.1, 2.2 and 2.3 primarily concentrate on part based analysis, in which a human skeleton is used to detect the functionality of the object part. In this work (Hu *et al.*, 2016), the object is conceptualized as a set of *proto-patches* that support multiple functionality. This is different from part based analysis in the sense that a specific region in an object part might support different functions. This is achieved by co-analysing typical object-to-object interactions and intra-object properties and relations. For example: The geometry of a table can allow a person to use it as a desk or sit on it. This is especially useful in synthesizing multi-purpose products (furnitures). This motivates us to learn a *functionality model* for a particular object category, e.g., strollers or backpacks, in which the model describes the functionality-related properties of the 3D objects belonging to that category. The model supports two important queries, *how*, via common interactions, and *why*, via patch localization, of functionality analysis, and learning the empty spaces around the objects which influence their functionality. The primary

ingenuity of this learning approach is that it helps in understanding the functionalities of new objects without a scene context.

2.4.1 General Overview

2.4.2 Functionality Modeling and Learning

The functionality model, as discussed above, is based on the idea of localized analysis of functionality. This is represented using patches, defined on the surface of the shapes, through which the geometric understanding of the underlying surface is revealed. The basis of the claim is that patches are more general than parts since, for example, the top and bottom of a part may support different functional properties. The functionality model is learnt by taking a central pivot object which is compared to all other objects in the scene, and represented with features derived from geometric entities, e.g., intersection bisectors and the interaction regions. The proto-patches are learnt by co-analysis of the patches derived from the interaction regions, in which we can easily establish a strong correspondence. Formally, a proto-patch is a patch prototype that is represented by a set of unary and functional space around it. The unary patches encode the point-wise geometric features. The model is denoted as a set of Proto patches, a set of binary patches and a weight vector which indicates the relevance of the unary and the binary properties in describing the functionality. The model stores a proto-patch for each of the functionality a object supports. This enables us to conceptualize in a high level abstract manner by listing the set of functionalities it supports. For example: A chair, which is typically used for sitting, might also be used as a desk.

2.5 Human Interaction Priors for Scene Design

We have seen in 2.1 and 2.2 that human interaction is very helpful in understanding the semantics of objects. This is exactly exploited by (Ma *et al.*, 2016) to evolve a scene based on the actions/interactions of the human with the objects in the scene. With the increasing need of indoor scene environments that comprise of many objects that we interact with frequently for many data driven analysis applications, this method for action-driven scene evolution aims to replicate how indoor scenes evolve in real life, producing continuous series of realistic virtual 3D scenes. An action can involve multiple humans and objects and alteration of scenes is through placement of object necessitated by actions.

2.5.1 Capturing the Environment in an Action Graph

The scene is evolved by transitioning in an action graph, essentially a Markov model formed over the action nodes which is developed by training over a large set of annotated 2D images. The action node is viewed as a set of the human pose, the action, the key object and priors trained for the occurrence probabilities of the objects as well as the spatial distribution of the objects in the scene w.r.t to the human and the other objects. Thus each scene is modeled as a human-object, object-object interaction set when a human is performing an action with a key object. The human pose is extracted from the 2D image by using existing methods to to find a 3D pose configuration whose projection on the image plane matches the 2D joint annotations. This pose is held in contention with a pre-decided set of poses for common indoor actions and the one with minimal deviated joint angles is chosen. The key objects are also pre-decided for indoor environments and extracted from the 2D-image annotations which also contain the action being performed. All the annotated objects are transformed into the object frame which has the human's right,frontal views as the x,y axes and the polar coordinates of the objects in this frame are used to model the spatial configurations later. The occurrences also capture multi-occurring objects within a scene through histograms.

2.5.2 Scene Evolution

The action graph is constructed very much with the idea that actions with similar semantics must have more linkage/transition probabilities. Semantic/Functional properties of a scene are the similarity in the key and the other objects occurring in the action. As we transition to a new scene i.e to a new action, objects are relocated and also may be inserted, so that the new interactions are in accordance with the new action. The spatial distribution priors modeled in training come to our aid when inserting a new object and also while relocating objects. Penalties physically implausible cases like collisions, objects falling off the edge are imposed. The non active objects are moved out of the working zone of the human pose. Scenes consisting of multiple humans can also be generated and evolved using the same method but also accounting for the interaction between the humans.

2.5.3 Implications

This method generated messy scenes at a very granular level and the reasoning is straight forward in a sense that the evolution is semantic-based that is human interactions drive the scene forward. This actually is similar to Shape2Pose where we imagined a

human to interact with a static object to obtain the characteristics of the object, where as now we have a human interacting with the environment of a dynamic scene and thereby the evolution of scene is guided by the interactions towards a realistic scene(realistic as the entire model is data driven). The plausibility and naturalness of thus obtained scenes are high when compared to the other scene generation methods that use just the scene data to evolve it and also compared to the scenes generated by an artist. This further stresses our point that the functional features of shapes are very much required and relevant for generation and development of objects and scenes containing them.

Chapter 3

Synthesizing Objects

3.1 Introduction

In this section, we present our work which is a result of ideas derived from all the aforementioned research papers. The ideas discussed in chapter 2, introduce us to two important concepts, the introduction of a hallucinating human actor in a scene as a cue complementary to visual appearance of the object. We infer that the functionality an object intends to support is dictated by it's geometric shape, and co-interaction of different parts of the human body with different parts of the object model. In addition to this, we deduce that through small deformations to an existing object, we can create new objects that better support a particular functionality. For example, a chair with a upright back rest can be a bit relaxing if the backrest is reclined. This motivates us to synthesize a furniture from scratch based on joint stress distribution in the human skeleton.

3.2 Algorithm Overview

We assume the human body is made up of rigid uniform thin rods, linked through joints which have flexibility. Given a pose of a skeleton, we try to design furniture that minimises the joint torques at different joints of the skeleton. The problem can be posed in many ways like minimising the maximal joint torque or minimising the overall average joint torque, or given prior data about thresholds on human joint torques, finding the minimal furniture that can be designed without collapsing the skeleton. For every formulation we require an algorithm to obtain estimates of joint torques at each joint of the skeleton. We achieve this through the following steps:

- Find the minimal number of ground supports and their locations so that the skeleton is at rest

- Calculate the values of the normal forces at the above supports
- Form the free body diagrams of different branches at a joint modeling the internal joint torques
- Calculate the values of internal joint torques at each joint for each branch of skeleton

After calculating the joint torques, we regulate the sampling density across the body by using the magnitude of stress components at the joints. For example: We want a higher sampling density to support our back because the surface area is more, and the bones / cartilages are weak. It's not very comfortable sitting without a back support or with minimal back support. Similarly, sitting on a pointed surface is also not desirable. On the other hand, it's quite common to have little or no head rest for chairs which support an upright sitting position, however, for a recliner seat, it's almost always desirable to have a head rest. These characteristic features of furniture models, which is tuned or designed to support a particular posture, motivate us to have a varying sampling density rather than uniformly distributed points across the whole body.

Our algorithm for raising the height field around the skeleton is as follows:

- Sample points on the Skeleton Link (or the region around it) directly proportional to the stress values at the two end joints of the particular link.
- Populate a point matrix $PM_{ij}|PM_{P_xP_y} = P_z, \forall P_{xyz}$.
- Plot a zero surface around the point cloud such that $Z_{ij} = 0$.
- Solve an optimization function using gradient descent algorithm by selectively pruning error values and refining height values.

3.3 Locating Minimal Supports

Since we require the pose of the skeleton to be unchanged, we restrict the motion of the joints, making them rigid hinges. So the entire skeleton can be viewed as a rigid body. Assuming k supports, we have a rigid body with k+1 forces on it including its weight with all force aligned along the z-axis(assuming gravity in the negative Z direction). We need to solve for the normal forces at the k points so that the skeleton remains at rest. From the Newton's second law, we arrive at the conclusion that the net external force on the skeleton is zero, thereby giving an equation in k variables. Since the skeleton is at rest w.r.t the ground any point on the skeleton or in the ground frame is inertial. This implies that the torque about any point in the inertial frame is equal and since the skeleton must

be at rest we equate it zero. Equating the net torque at any point to zero gives one more independent equation in the k variables. Writing the equation for equating net torque to zero at any point can be shown as a linear combination of the following equations :
(i) Net external force on the skeleton is zero
(ii)Net torque about the center of mass of the skeleton is zero. Therefore we have two equations with k unknowns.

$$N_1 + N_2 + ... + N_k = mg \tag{3.1}$$

$$x_1N_1 + x_2N_2 + ... + x_kN_k = 0 \tag{3.2}$$

where x_p is the norm of XY-component of the position vector of support point p w.r.t the center of mass of skeleton. Therefore the system is under-determined if k≤2. So minimal points to get the skeleton to rest is one only if there exists a support point with $x_p=0$, otherwise we can find two such points.

3.3.1 Finding the two minimal points

Taking k=2, we see from (3.2) that x_1 and x_2 are collinear, therefore x_1,x_2 and the center of mass lie in the same plane perpendicular to the XY-plane. Also the support points can be seen as an (x,y) pair as the (x,y) pair determines z as the lowest point(least z coordinate) on the skeleton with the same x,y values. Therefore it reduces to finding two points lying collinear with the center of mass in the 2-d projection of the skeleton on the x-y plane.

Now we have determined all the external forces on the body and so now the internal joint torques are deterministic

3.4 Free Body Diagrams at each Joint

We model the internal joint torque as a couple caused by internal forces that actually don't cause any net force but create an internal joint torque for each branch of the skeleton arising from the joint. To be specific, perform cuts at every outgoing bone from the joint and for each cut draw the free body diagram with all the previous forces with an additional joint torque at the cut. When seen in the free body diagram of the joint without any cuts, the sum of these internal joint torques is zero thereby there is no net internal torque generated, but on each cut free body diagram there can be a finite internal joint torque acting at the cut point. For each free body diagram, equate the net torque to zero and solve the equation to find the value of the only variable i.e the internal joint torque acting on this branch. This implies there are multiple internal joint torques at a k-branch

joint T_1,T_2,...,T_k which add up to zero internal torque. We define the estimate of joint torque at this point as the $max\{T_1,T_2,...,T_k\}$

3.4.1 Solving the cut-FBD

Given the cut-FBD, we know the external normal forces acting on it, also the weight of the bones in that branch form an external force. The internal joint torque at the cut is equal to the net torque to to all other external forces. To collect the weight of the bones in that branch we perform a dfs from the joint neighbor to our working joint with our joint marked as visited.

3.5 Sampling the Human Skeleton

Given the human skeleton, the above discussed approach gives us the stress values at each joint. Next, we sample points on the skeleton proportional to the average of the stress values at the terminal ends of a skeleton link. In order to sample points around the human body, we assume that the link has a rectangular surface area around it, where width of the rectangle is assumed to be one-half of it's length.

3.6 The Optimization Setup

Once we have a set of points on the *human skeleton*, the next objective is to raise a height field, so as to reduce the joint stress values. For example: In a sitting posture, the human skeleton has a very high stress value at the knee joint, and the pelvic joint. This intuitively guides a designer in putting a seat and a backrest in a *sittable* furniture. In our approach, we model the furniture as a height field, which is essentially a 2D matrix with height values encoded in it. The points are also distributed in the 3D space, and therefore, it is also encoded as a 2D matrix (which lies above the height field matrix), and stores of z values of the points in the position (x, y).

In order to avoid strict fitting with the points and form a smooth curve, we penalise the *square of the laplacian* (Nealen *et al.*, 2006) at all points. This is based on observation that, a person sits comfortably on a flat surface rather than a pointed surface e.g., a traffic cone. In addition to this, we also incorporate an additional penalty, *intersection penalty* which heavily penalises the points on the height field which lie above the skeleton. This is fairly simple, because a skeleton must strictly rest on top of the furniture, therefore, intuitively, it's equivalent to going over the height field one at a time, and dragging down points which have a height value greater than the point (on the skeleton) lying above it.

Mathematically, we setup the optimization problem as follows:

$$\min\left(\alpha * ||PM - H||^2 + (1-\alpha) * \frac{||Conv(H)||^2}{2} + C * InterPen\right) \tag{3.3}$$

α, C are the hyper-parameters that have been tweaked in the context of this problem. In addition to this, we have used varying weights for the points of the height field. As illustrated in the figures, this allows the surface to raise up freely which otherwise gets stuck, because the points which have 0 height values and never rise up pull down the surface way too much. Another plot that shows the importance of tuning the parameter α has been illustrated in figure 4.1.

3.6.1 Solving the Optimization Problem

In order the calculate the laplacian at all points, we use a 2D convolution filter as follows:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & +8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

This filter is an edge detection filter in images and therefore, penalizes sharp difference in heights. This is a very common filter used to calculate the laplacian.

Next, we use gradient descent to solve the problem. For gradient descent we do the following:

```
while iterations < MAXITER AND error >= precision
    $H^' = H - lambda * grad(f)$
    $H = H'$
end loop
```

where, H is the height field, *lambda* is the learning rate, error is taken as the square of the euclidean distance between the height field and the point field, and $grad(f)$ is the gradient of the optimization function f, which is defined as follows:

$$\alpha * -2 * (PM - H) + (1-\alpha) * Conv(Conv(H)) + C * InterPen \tag{3.4}$$

The intersection penalty matrix is calculated using an activation function that gets activated when $PM - H$ is positive. Ideally, we want a setup where the activation is a binary activation function based on whether the skeleton intersects the surface at point (x, y). However, the gradient of such a function is not defined at all points, therefore, for

the purpose of mathematical correctness, we use *sigmoid* activation function which has a continuous first derivative. Moving forwards, we want to verify whether the choice of activation actually makes a difference in correct generation of the height field.

3.7 Ongoing Work

Till now given a skeleton at rest using minimal supports, we find the joint torques and sample points accordingly to generate the furniture. Now instead of finding the torques just once in the beginning, we provide an algorithm to iteratively provide supports in the order of their importance. Once a support has been added, torques are recalculated in order to calculate the next support. So the problem changes to finding the support that causes maximum reduction in the joint torque energy function, given the skeleton topology and the supports already present. The skeleton configuration can be densely modeled as $S(B, J, C, N, d)$ where (i) B denotes the bones (ii) J enlists the joints (iii) C captures the connectivity of the bones and the joints (iv) N contains the supports to the skeleton (v) density of the uniform mass skeleton. Given a skeleton configuration the joint torque energy function can be modeled as either the aggregate of all the joint torques or the maximum joint torque. Given S with $k-1$ supports, we require to find the kth that produces the maximum reduction in the energy function. Assume the new support $\mathrm{N_k}$ is added at position $\mathrm{P_k}$.

$$P_k = \lambda P_{\mathrm{j_a}} + (1-\lambda) P_{\mathrm{j_b}} \quad \lambda \in [0,1]\,, (j_a, j_b) \in B \tag{3.5}$$

The problem reduces to:

$$\arg\min_{b \in B} \left(\arg\min_{\lambda} E(S(B, J, C, N \cup N_k^b, d)) - E(S(B, J, C, N, d)) \right) \tag{3.6}$$

subject to the constraints in section 3.3.

Chapter 4

Results and Discussions

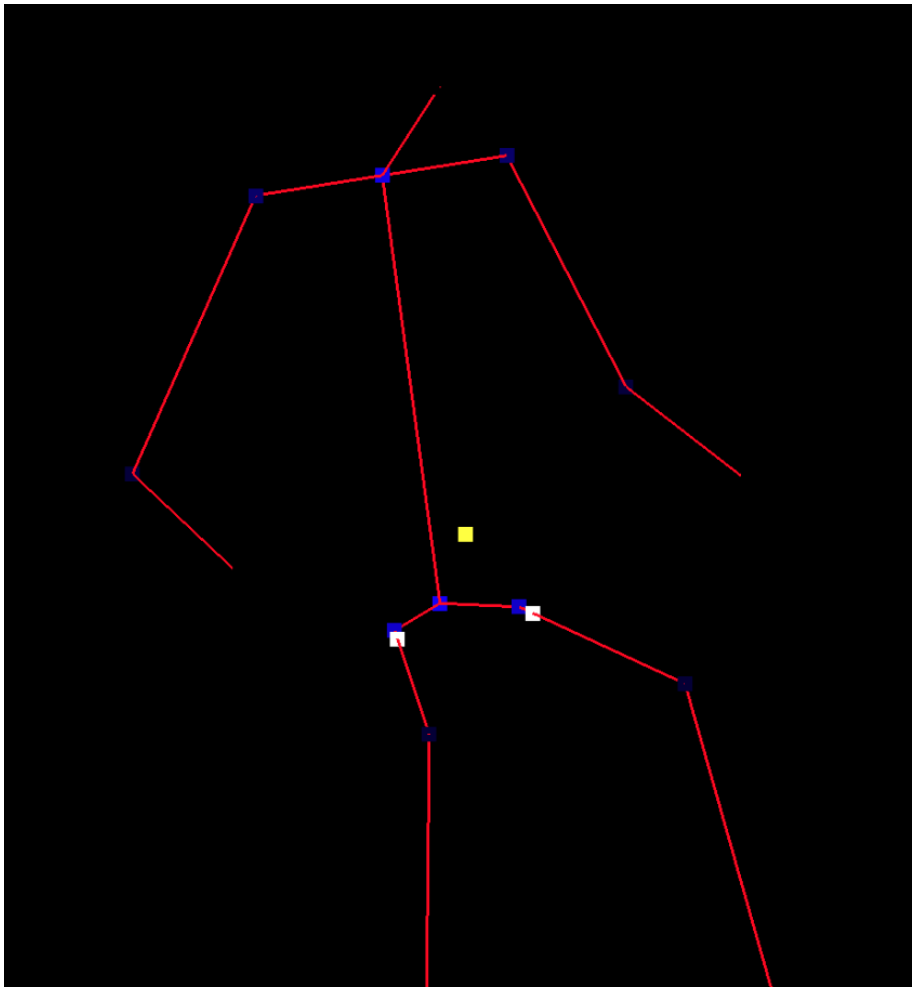
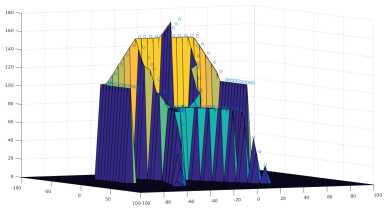
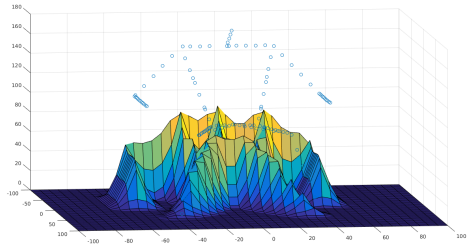


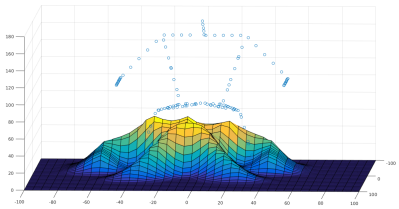
Figure 4.1: The yellow point is the center of Mass and the two white points represent the two minimal supports. The intensity of the blue color at each joint captures the maximal internal torque is joint is bearing.



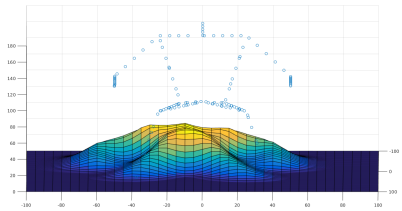
(a) alpha = 0



(b) alpha = 0.1



(a) alpha = 0.5



(b) alpha = 0.7

Figure 4.1: Visualisation of the tuning of the value of α which essentially governs whether the smoothness factor is given more weight or the surface lift-up term.

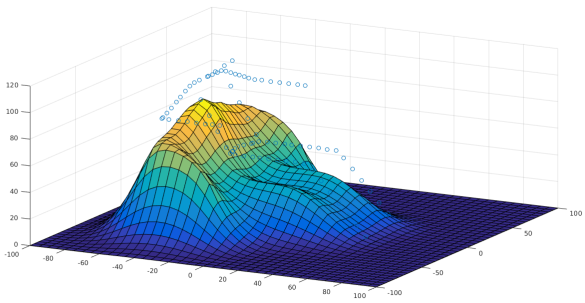


Figure 4.2: Surface generated using uniform sampling. Notice how there are ridges created when we have a linear sampling. This motivates us to do a 3D sampling over a realistic model of human body.

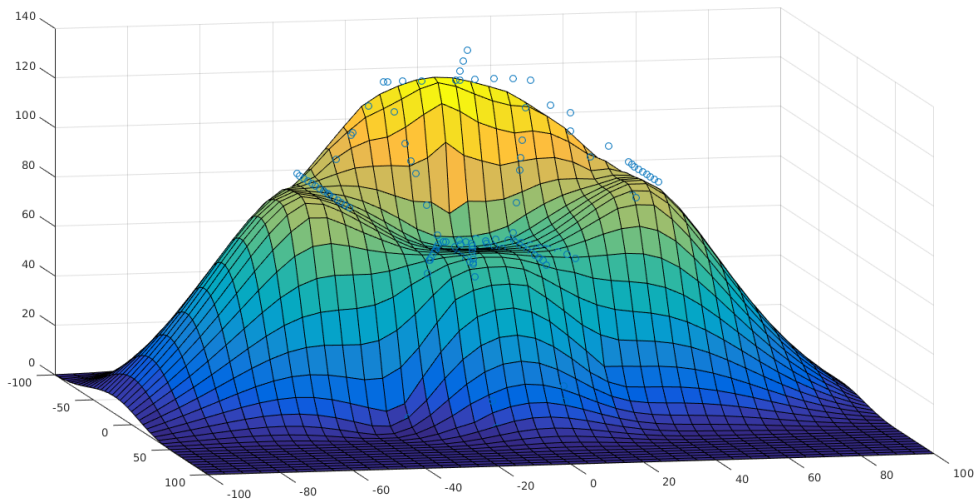


Figure 4.3: Automatically generated chair (sofa) using human pose (skeleton orientation) as input. The surface that is generated does not assume any statistical prior on "how humans sit", i.e., the algorithm has zero knowledge about which part of the body is to be supported so that a particular pose can be comfortably supported. The points on the body are sampled proportional to the magnitude of joint torques. The overall objective is to sample more points near the joints with maximum torque values and make the person sit on the 3D shape that is generated.

4.1 Future Work

Presently, we are able to successfully synthesize a furniture for the sitting posture. In order to further extend our work, we finally want to generate a single furniture that supports multiple postures. For example, a person can sit and sleep on a fairly big sofa. This problem becomes even more challenging because it necessitates the incorporation of an additional constraints like surface area, material use etc. In addition to this, we need to formulate our sampling and stress model in a better way so that it correctly captures the regions of the human body which need to be supported with an underlying surface.

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